

BRAT: Bidirectional Relative Positional Attention Transformer for Event-based Eye tracking

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Abstract

Event-based eye tracking, with its neuromorphic triggering mechanism, offers high temporal resolution and low power consumption, making it a promising technology for future application. However, this triggering mechanism is a double-edged sword. While providing exceptional temporal precision for eye tracking, it also introduces challenges such as the loss of static information, increased complexity of dynamic signals, and irregular spatio-temporal distribution of events. To tackle these challenges, this paper presents a Bidirectional Relative Positional Attention Transformer (BRAT) architecture, designed to fully exploit the spatio-temporal sequence information within event streams, enabling stable and precise eye tracking. The proposed network is composed of a spatial encoder and a temporal decoder. The former utilizes a CNN to extract geometric structural features from event representations, while the latter combines a Bi-GRU block and the BRAT block to analyze temporal motion patterns and accurately localize pupil positions. Furthermore, we propose a multi-time-step training strategy, which improves the model's stability and accuracy by incorporating event representations across multiple time spans as input. Tests on the ThreeET-plus benchmark demonstrate BRAT's high tracking accuracy and stability for complex eye movement patterns, and achieving state-of-the-art performance.

1. Introduction

Eye tracking aims to localize the pupil position over time, enabling accurate estimation of visual attention. It plays a critical role in various domains, such as behavioral analysis in psychology [21, 24], human-computer interaction in AR/VR systems [1], and gaze-contingent rendering [2, 16].

Traditional eye-tracking methods often rely on costly high-speed cameras, as eye movements like saccades and

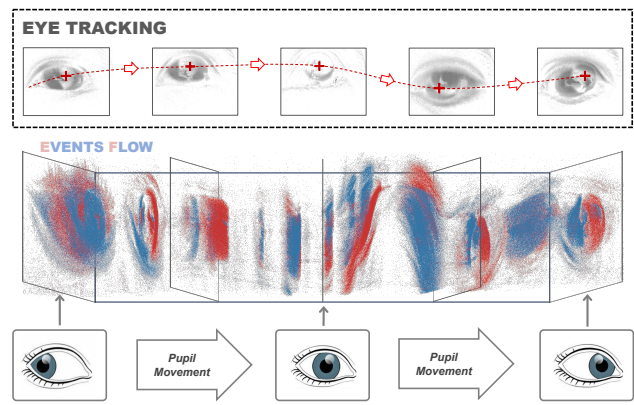


Figure 1. A schematic representation of event-based eye tracking. The asynchronously triggered event camera excels at capturing the dynamic motion of the pupil, while the tailored network architecture accurately extracts its position.

blinks occur rapidly and require high temporal resolution for accurate capture. However, such systems suffer from substantial data storage demands and limited potential for lightweight deployment. In contrast, the introduction of event cameras [12] can effectively address this issue. Due to their asynchronous triggering characteristics, they enable high temporal resolution, low-power consumption, and long-duration collection of eye movement processes.

However, the unique spatio-temporal properties of events brings challenges to eye-tracking tasks, which can be broadly categorized into two main aspects: Firstly, the absence of static information. Human eye movements often involve prolonged periods of gaze fixation or rest states. Since event cameras rely on motion-triggered sampling, gaze fixation generates minimal dynamic information, making it difficult for the network to accurately extract the pupil's position. Secondly, blinking or eye closure can generate a large volume of event signals, yet the pupil remains occluded, making it challenging to track effectively. The above factors combined make it difficult to effectively ex-

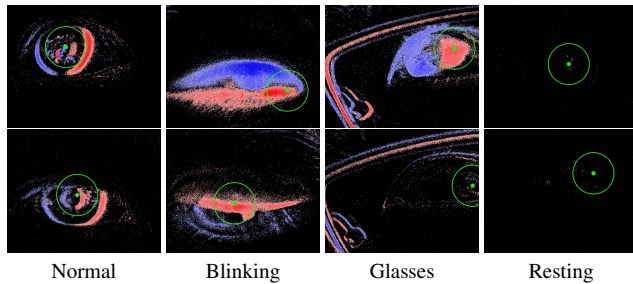


Figure 2. The figures showcase challenging examples in event-based eye-tracking, such as blinking, interference from wearing glasses, prolonged fixation and resting. Specifically, the dot represents the ground truth position of the pupil, while the circle indicates a 10-pixel Euclidean distance range around the GT.

tract eye movement information from the irregularly distributed event streams in time and space.

To address these challenges, incorporating continuity priors and leveraging motion context information for precise tracking is a natural approach. Previous works have explored this by introducing Recurrent Neural Network (RNN) [8] to extract eye movement sequences or employing Spiking Neural Network (SNN) [5] to asynchronously process event streams. Additionally, the Mamba architecture [13, 27] has been utilized to achieve more stable tracking and improved model generalization. However, prior approaches have struggled to explicitly model directional temporal dependencies of pupil movement, limiting their capacity to capture fine-grained event dynamics.

In order to achieve fine-grained perception of pupil position within the motion context, we propose **BRAT**, a **Bidirectional Relative Positional Attention Transformer**, which enables precise and stable pupil localization from event signals with uneven spatiotemporal distribution. It also maintains robust tracking performance even under challenging conditions such as gaze fixation, blinking, and fast saccades. Specifically, the event-based eye-tracking architecture is composed of a spatial encoder and a temporal decoder. The spatial encoder leverages a CNN to extract geometric structural information from events, which is then passed to the temporal decoder. Where temporal sequence features are initially extracted using a bidirectional GRU and subsequently fed into the BRAT block for precise perception of eye movement patterns. Furthermore, to improve the model’s generalization ability, a multi-time-step training strategy is proposed, which combines event representation sequences with different time steps as input. Experiments on challenging eye-tracking datasets demonstrate that this approach effectively enhances tracking stability and accuracy.

In summary, our contributions are as follows.

- A novel framework named BRAT, is crafted to explicitly

model directional temporal dependencies in complex and abrupt pupil movement sequences.

- A multi-time-step training strategy is proposed, leveraging input event sequences across varying time steps to train the network, ensuring robust performance.
- The proposed method achieves superior performance on exceptionally tough eye-tracking tasks, and won the championship in the 3ET challenge at the CVPR Event-based Vision Workshop 2025 [9].

2. Related Work

2.1. Eye Tracking

Eye tracking task aims to achieve real-time monitoring of pupil positions. After years of development, various approaches have emerged. Among them are hardware-based methods such as Electrooculography (EOG) [3, 17], which determines gaze direction by measuring the resting potential of the eye, and scleral search coil technology [20, 28], which tracks the direction of the eye by following contact lenses worn by the user. However, these methods rely on complex hardware devices and require contact-based setups, limiting their application scenarios. Currently, the most widely used approach is video oculography (VOG) [6, 7, 14, 15, 18, 22, 29], which tracks the pupil by analyzing the eye movement process captured by a camera through algorithms. However, due to the high-speed and sudden nature of eye movements, traditional frame-based cameras face limitations in sampling rates, making it challenging to achieve high-speed real-time tracking. In contrast, event-based methods, with their asynchronous sampling strategy, allow for long-term sampling of targets with low data burden and low power consumption. These methods show great potential for development in eye-tracking tasks. Early model-based methods determined the 3D parameters of eye pose by fitting event information into images. However, these methods still rely on image data.

Recent advancements in purely event-based eye-tracking methods can be divided into two main approaches. One focuses on directly extracting eye movement features from raw event streams, such as SNN-based methods [5]. The other extracts features from 2D event representations sampled at fixed intervals, with notable examples including RNN-based [8] and Mamba-based methods [27]. The first approach enables lossless information extraction directly from the event stream. However, due to being in a nascent stage, it has yet to demonstrate dominance in the field. In contrast, methods based on 2D event representation sequences have been successfully applied to eye-tracking tasks. However, they primarily focus on modeling local temporal relationships, which limits their performance in cases involving prolonged fixation or high-frequency blinking. These challenges call for methods capable of explicitly

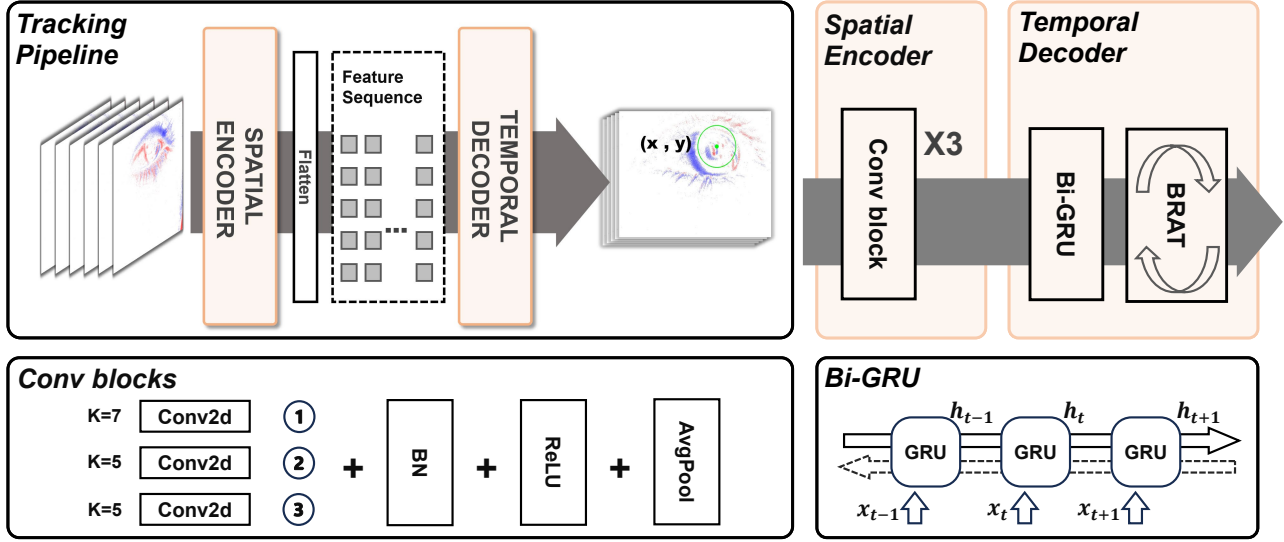


Figure 3. **TOP:** BRAT pipeline. The method first captures spatial context using large-kernel convolutions, followed by sequential modeling of local and global temporal dependencies via a bidirectional GRU and a transformer with bidirectional relative positional attention. **Bottom:** The three large-kernel convolutional layers, followed by pooling layers, progressively enlarge the receptive field for the feature maps. The bidirectional GRU captures local temporal dependencies.

modeling the full-sequence temporal dependencies of pupil movement. In this paper, we introduce a novel transformer architecture with bidirectional temporal relative positional encoding, designed to perceive the entire event sequence. Additionally, we incorporate a unique training strategy to further enhance the model’s performance.

2.2. Relative Positional Attention

Transformers [25], unlike recurrent architectures, lack a recursive structure to inherently modeling sequential dependencies. As a result, positional encodings are essential for capturing temporal relationships within sequences. Vanilla absolute positional encoding uses sinusoidal functions to assign fixed positional embeddings to sequence elements. However, this method fails to explicitly represent relative positional relationships between elements and exhibits poor generalization in tasks with irregular temporal distributions, such as eye-tracking. Relative positional encoding [4, 10, 11, 19, 23], on the other hand, excels at capturing the relative temporal dynamics of movements, offering superior generalization abilities. Furthermore, it demonstrates robust performance in challenging eye tracking scenarios such as blinking or resting, where local temporal context is critical for motion representation. To further enhance the modeling of temporal dependencies, we introduce a novel bidirectional relative positional attention transformer (BRAT) architecture. This approach enables fine-grained capture of both forward and backward dependencies within eye movement sequences. Specifically, the BRAT module is integrated to refine temporal motion fea-

tures extracted by the Bi-GRU module, achieving state-of-the-art results in eye-tracking tasks.

3. Methodology

In this section, we will introduce our methodology. Sect. 3.1 provides a detailed explanation of our network architecture. And we will discuss the event data processing strategy in Sect. 3.2. Finally, the supervision strategy is presented in Sect. 3.3.

3.1. Architecture of Proposed Network

The network architecture consists of a spatial encoder, which extracts geometric information from each event representation, and a temporal decoder to capture the temporal relationship of eye movement sequences. As shown in Fig. 3, the spatial encoder is composed of a series of Convolution blocks, while the temporal decoder is constructed by cascading a Bi-GRU block and the BRAT block.

Spatial Encoder. In the spatial feature extraction phase, the input binary representation is first passed through a convolution layer with a kernel size of 3, producing 32 channels. The spatial features are then extracted using three subsequent convolutional layers with larger kernel sizes of 7, 5, and 5. The use of larger convolutional kernels is motivated by the fact that the pupil’s position is strongly correlated with the positions of other facial features, such as the eye corners and eyebrows. Larger kernels, such as those with sizes 7×7 and 5×5 , provide a larger receptive field, which

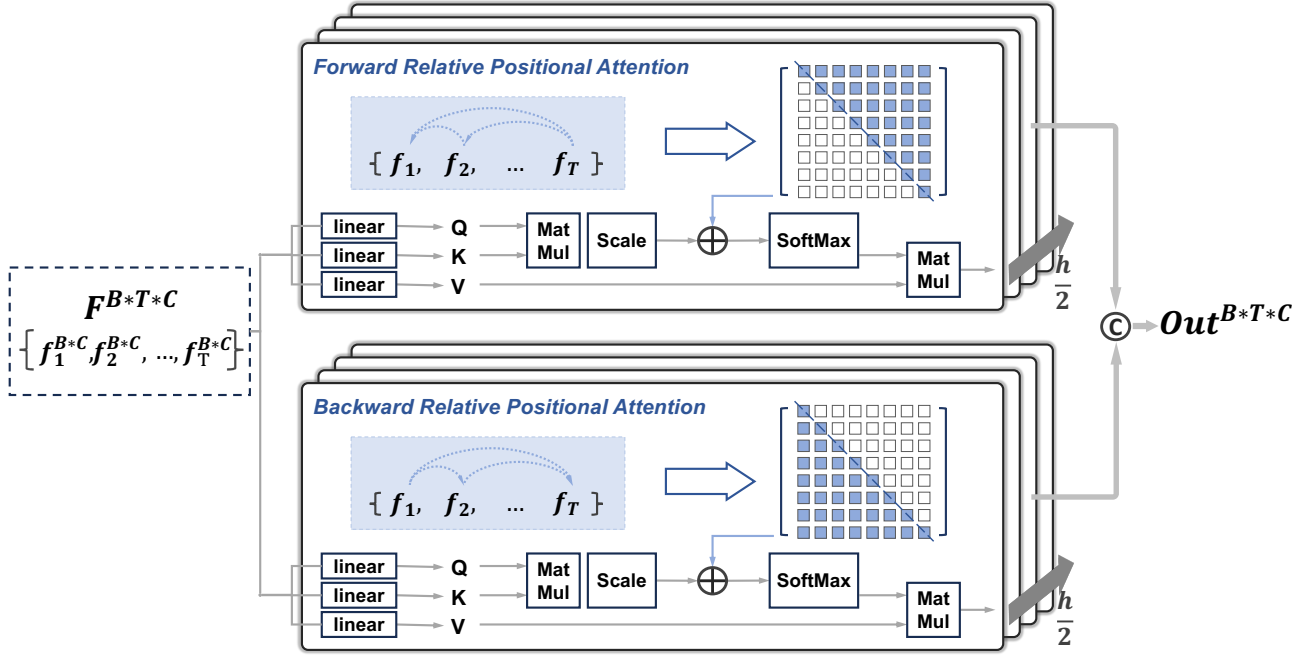


Figure 4. Bidirectional Relative Positional Attention. The attention mechanism utilizes two sets of attention heads to compute the forward and backward attention matrices separately. By decoupling the relative position encoding, each branch focuses on distinct temporal dependencies, effectively disentangling the previously mixed influences of past and future contexts.

allows the network to capture additional spatial context. This enhanced spatial perception often leads to improved auxiliary localization accuracy. After the first two large-kernel convolutions, pooling layers are applied to reduce the spatial dimensions. Following the third convolutional layer, the feature maps undergo a spatial dropout layer to mitigate overfitting. Unlike standard dropout, which randomly drops individual elements, spatial dropout drops entire rows with a certain probability. This operation helps to simulate scenarios where parts of the pupil are occluded while maintaining the spatial structure of the feature map. In contrast, standard dropout, which randomly drops individual pixels, could disrupt the spatial relationships that are critical for accurate pupil localization.

Bi-GRU. The bidirectional GRU captures temporal dependencies by processing the sequence in both forward and backward directions. It computes update and reset gates based on the current input x_t and previous hidden state h_{t-1} , controlling the flow of information and allowing the model to adjust memory retention. Unlike LSTMs, GRUs are computationally more efficient and easier to train, but a unidirectional approach may struggle with context at the beginning or end of sequences. By processing input both ways, the bidirectional GRU ensures better coverage of the temporal context.

BRAT. While the bidirectional GRU effectively captures short-term temporal patterns, it remains limited in modeling long-range dependencies due to its sequential nature. This becomes critical in complex eye movements such as “smooth pursuit”, where pupil motion evolves gradually over extended periods, and “blinks”, where transient occlusions require contextual inference from both past and future frames. To overcome these limitations, we incorporate a transformer-based bidirectional attention module with relative positional encoding. This component complements the GRU by enabling global temporal reasoning, allowing the model to capture long-range interactions and refine predictions under challenging motion dynamics.

Specifically, the standard multi-head self-attention in transformer is replaced with a bidirectional variant that explicitly encodes relative temporal distances, see Fig. 4. Given a sequence of length T and h attention heads, the output of each head is computed as:

$$\mathbf{Attention}^i = \text{softmax} \left(\frac{\mathbf{Q}^i \mathbf{K}^{i\top}}{\sqrt{d_k}} + \mathbf{B}^i \right) \mathbf{V}^i, \quad (1)$$

where $\mathbf{B}^i \in \mathbb{R}^{T \times T}$ is a relative position bias matrix designed to modulate attention weights based on temporal distance. It is decomposed into forward and backward compo-

nents as:

$$\mathbf{B}_{forward}^i = \begin{cases} m^i \cdot (t - s), & t \geq s \\ 0, & t < s \end{cases}, \quad (2)$$

$$\mathbf{B}_{backward}^i = \begin{cases} 0, & t \geq s \\ m^i \cdot (s - t), & t < s \end{cases}, \quad (3)$$

where m^i denotes the sensitivity of head i to the relative position, generated via a monotonically decreasing linear mapping to progressively diminish attention to distant steps. This formulation allows each head to focus on either past or future steps, capturing asymmetric temporal dependencies. By decoupling the forward and backward attention mechanisms, the model independently models past and future contexts, improving its ability to represent complex temporal dynamics.

3.2. Event Data Processing

Event Representation. An event is triggered when the logarithmic light intensity on a pixel of the event camera sensor $\log \mathbf{I}$ exceeds the threshold c :

$$|\log \mathbf{I}(x_i, y_i, t_i) - \log \mathbf{I}(x_i, y_i, t_i - \Delta t_i)| \geq c. \quad (4)$$

And we define each event as

$$e_i := (x_i, y_i, t_i, p_i), \quad (5)$$

where t_i is the triggering time, (x_i, y_i) are the pixel coordinates, and $p_i \in \{-1, 1\}$ is the triggering polarity.

When the logarithmic difference is positive, the polarity is 1; when the logarithmic difference is negative, the polarity is -1 . Based on the above mechanism, motion and illumination changes in the scene trigger a spatiotemporal event stream. In our method, the event stream within each fixed time interval is integrated into a 2D feature representation, and the sequence of feature representations is used as the network input.

Multi-time-step training strategy. It is worth mentioning that we adopted a multi-time-step data sampling strategy during training, which effectively improved the robustness of the model. Specifically, a sliding window with a fixed length was applied over the long event sequence with a stride of 1, generating dense training samples. Within each window, frames were sampled at uniform intervals defined by a step size, allowing the model to observe motion over longer temporal spans while controlling input density. The implementation strategy is as follows. Unlike traditional sequential inputs, we sampled the event representation sequence at equal time intervals, as shown in Fig. 5. Instead of using sequences like

$$R_1, R_2, R_3, \dots, R_T, \quad (6)$$

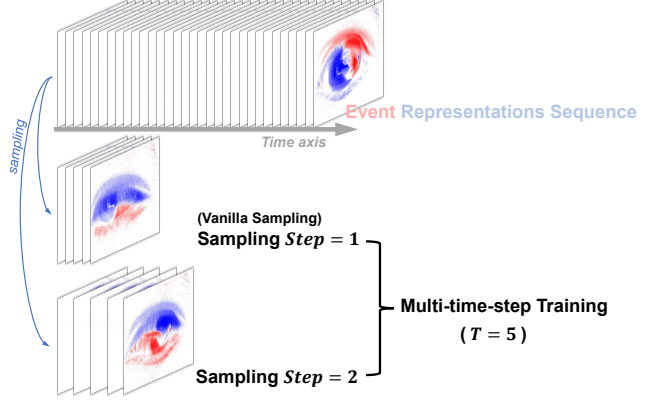


Figure 5. A schematic illustration of multi-time-step training. For simplicity, the input sequence length is set to $T = 5$ for demonstration purposes. In practice, the $T = 45$ and training is performed using a mix of data sampled with $Step$ from 1 to 5.

we expand them to

$$R_{1*Step}, R_{2*Step}, R_{3*Step}, \dots, R_{T*Step}, \quad (7)$$

where R_i represents the event representations, $Step$ is a fixed integer, and T denotes the input sample length for the model.

This strategy brings two benefits. Firstly, by selectively disregarding certain intermediate motion information, it effectively simulates the dynamics of rapid saccadic eye movements, enabling the model to achieve enhanced tracking accuracy under such challenging scenarios. Additionally, this approach is equivalent to increasing the number of training samples, enhancing the robustness of the model.

Data Augmentation methods. Unlike image-based eye-tracking tasks, where data is relatively easy to collect, datasets for event-based eye-tracking tasks are scarce, limiting the generalization ability of models. Therefore, we adopted a series of data augmentation strategies to improve the model's generalization ability and accuracy, including spatial translation, horizontal flipping, temporal shifting, and event cutout [27]. Vertical flipping is excluded, as it does not align with real-world eye movement patterns. Temporal shifting refers to circularly shifting the input sequence forward or backward, such that a sequence like $[0, 1, 2, \dots, T]$ might become $[2, 3, \dots, T, 0, 1]$ or $[T-1, T, 0, 1, \dots, T-2]$. This augmentation introduces temporal diversity, enabling the model to learn more robust representations invariant to absolute temporal positions.

3.3. Supervision

The loss is computed by first calculating the squared differences between the predicted values and true labels at

Method	p_5	p_{10}	p_{15}	p_{error}
CNN-GRU [9]	0.802	0.945	0.970	4.46
CB-ConvLSTM [8]	0.869	0.970	0.982	3.41
MambaPupil [27]	0.963	0.990	0.997	1.34
Ours	0.978	0.995	1.000	1.14

Table 1. Comparison with other methods: p_n denotes prediction accuracy within n pixels (higher is better), while p_{error} indicates the average Euclidean distance (lower is better). All models were retrained using the proposed multi-time-step training strategy.

each time step, followed by applying the square root and averaging over the time dimension:

$$Loss = \frac{1}{T} \sqrt{\sum_{t=1}^T (y_{t,pred} - y_{t,label})^2}. \quad (8)$$

This formulation captures the overall error across the temporal sequence, normalizing by the sequence length to account for varying durations.

4. Experiments

4.1. Experimental settings

Dataset and metrics. Training and evaluation were conducted on the ThreeET-plus dataset [9, 26], which contains 13 recorded scenarios, each comprising 2 to 6 segments with an event resolution of 640×480. The captured eye movements are broadly categorized into five types: random movement, saccades, reading text, blinking, and smooth pursuit. Pupil positions were annotated at a frequency of 100Hz. Following the protocol in [27], 75% of the data was used for training and 25% for validation. The predictions are generated at a frequency of 100Hz, matching the ground truth. Evaluation was performed using the Euclidean distance between predicted and ground truth pupil centers, as well as the success rate under pixel thresholds p_5 , p_{10} , and p_{15} .

Implementation details. The proposed method was implemented in PyTorch and trained on four NVIDIA RTX2080Ti GPUs. Training was conducted for 800 epochs with a batch size of 32. All networks employed the Adam optimizer and the Cosine Annealing Warm Restart scheduler, with an initial learning rate of 0.001. Each training sequence had a length of 45 and was sampled with a stride of 1 and step sizes ranging from 1 to 5. The choice of step significantly influenced performance—larger steps yielded more challenging sequences, while smaller steps increased the number of training samples. All inputs were downsampled to a resolution of 80×60, and ground truth labels were normalized accordingly.

4.2. Quantitative results.

The proposed method is evaluated on the ThreeET-plus dataset against several strong baselines, including CNN-GRU [9], CB-ConvLSTM [8], and the recent state-of-the-art MambaPupil [27]. To ensure a fair comparison, all models are retrained using the same multi-time-step training strategy described in Sec. 3.2.

As summarized in Tab. 1, the proposed method consistently outperforms all baselines across all evaluation metrics. Specifically, it achieves a significant improvement in p_5 , p_{10} , and p_{15} , while simultaneously reducing the average Euclidean prediction error p_{error} to 1.14 pixels. Compared to CNN-GRU and CB-ConvLSTM, which primarily model unidirectional or short-range temporal dependencies, the proposed approach better captures complex motion patterns and long-range correlations.

Notably, while MambaPupil introduces bidirectional GRU modules for enhanced temporal modeling, it remains limited to local sequential patterns and lacks the ability to capture global temporal context. In contrast, The proposed method leverages the bidirectional relative position attention mechanism, which explicitly encodes relative temporal positions and complements the GRU modeling. This synergy enables the model to maintain temporal consistency over time and better handle challenging scenarios such as saccade and occlusion.

4.3. Ablation study.

This section investigates the impact of various factors on model performance, including the choice of convolutional kernel sizes, the integration of the BRVT module, and different sampling step configurations.

Kernel sizes & BRAT. Table 2 presents the results of an ablation study that investigates the impact of different convolutional kernel sizes and the inclusion of the Bidirectional Relative Positional Attention Transformer (BRAT) mechanism on the model’s performance. The kernel sizes vary across three layers, and each configuration is tested with and without the BRAT mechanism. “Without BRAT” indicates configurations where the BRAT mechanism was excluded, and the network directly outputs through the GRU without

Kernel size	BRAT	p_5	p_{10}	p_{error}
(3,3,3)	✗	0.909	0.935	2.32
(5,3,3)	✗	0.913	0.948	2.25
(7,3,3)	✗	0.925	0.964	2.16
(7,5,3)	✗	0.932	0.978	2.14
(7,5,5)	✗	0.936	0.981	2.12
(7,7,5)	✗	0.928	0.963	2.20
(7,7,7)	✗	0.924	0.973	2.17
(7,5,5)	✓	0.954	0.979	1.99

Table 2. Ablation study across different structures. “(7,5,3)” represents convolutional kernel sizes of 7, 5, and 3 for three layers, with similar notation for other configurations. “w/o BRAT” indicates the network directly outputs through GRU without the BRAT mechanism. For experimental efficiency, all experiments are conducted with a sampling step of 5 and a stride of 1.

the additional attention mechanism.

The results demonstrate a clear performance trend: as the kernel size increases, the model shows continuous improvement in both the p_5 and p_{10} accuracy metrics, as well as a reduction in the average Euclidean error, p_{error} . However, this trend does not persist indefinitely. The configuration with a kernel size of (7,5,5) achieves the optimal performance, with the highest p_{10} accuracy of 0.981 and a reduced p_{error} of 2.12. The introduction of BRAT leads to a significant reduction in p_{error} , indicating that the bidirectional relative positional attention effectively captures long-term temporal dependencies and enforces better temporal consistency in the predicted points.

This ablation study emphasizes the importance of both convolutional kernel size and the BRAT mechanism in improving model performance. The use of larger kernels improves feature extraction from spatial inputs, while the BRAT mechanism enhances the model’s ability to capture bidirectional temporal dependencies, leading to better overall performance.

Sampling steps. This ablation study systematically investigates the impact of different temporal sampling steps on model performance, as presented in Tab. 3. Each row corresponds to training conducted using sequences generated with a specific sampling step, allowing us to isolate the effect of temporal granularity on model behavior. The “Mixed” setting refers to training with a combination of all sampling steps, providing a diverse temporal context.

The results reveal that smaller sampling steps (e.g., step=1) generally lead to higher prediction accuracy within short distances (as reflected in p_5), which can be attributed to the dense temporal resolution offering rich motion infor-

Step	p_5	p_{10}	p_{error}
1	0.984	0.994	1.19
2	0.974	0.990	1.24
3	0.977	0.989	1.44
4	0.978	0.975	1.67
5	0.954	0.979	1.99
Mixed	0.978	0.995	1.14

Table 3. Ablation study with different sampling steps. Each row represents training using sequences constructed with a specific sampling step. “Mixed” indicates training with all sampling steps combined.

mation. However, such high-frequency sampling may pose challenges for capturing long-range dependencies and may limit generalization in scenarios involving complex temporal dynamics. As the sampling step increases, the model benefits from observing longer temporal spans, which aids in capturing coarse-grained motion patterns. However, this also results in sparser input sequences, leading to a degradation in fine-grained prediction accuracy and a noticeable increase in the average Euclidean error (p_{error}), particularly for steps 4 and 5.

Notably, the “Mixed” configuration achieves the best overall performance, especially in terms of p_{10} and p_{error} , indicating that multi-scale temporal sampling offers complementary information. By exposing the model to diverse temporal resolutions during training, it becomes capable of simultaneously learning fine-scale motion details and capturing long-term temporal structures, resulting in more robust tracking performance.

4.4. Qualitative Comparison

In this section, we conduct qualitative comparisons of different mainstream event-based eye-tracking methods, as shown in Fig. 6. And we evaluate models’ performance across five scenarios: **Normal case:** A case where eye movements are clearly captured, serving as a baseline. **Fast saccade:** A high-speed scenario involving rapid eye saccade, demonstrating the high temporal resolution of event-based methods. **Subject wearing glasses:** A case highlighting the model’s robustness to redundant interference signals caused by glasses. **Eye Blinking:** A challenging scenario where frequent blinking temporarily occludes the pupil, making it impossible to extract its position from the current events representation. **Fixation/Eye resting:** A case with no triggering events, where the absence of information also makes pupil localization difficult.

It can be observed that BRAT effectively handles these challenging scenarios by precisely encoding the contextual

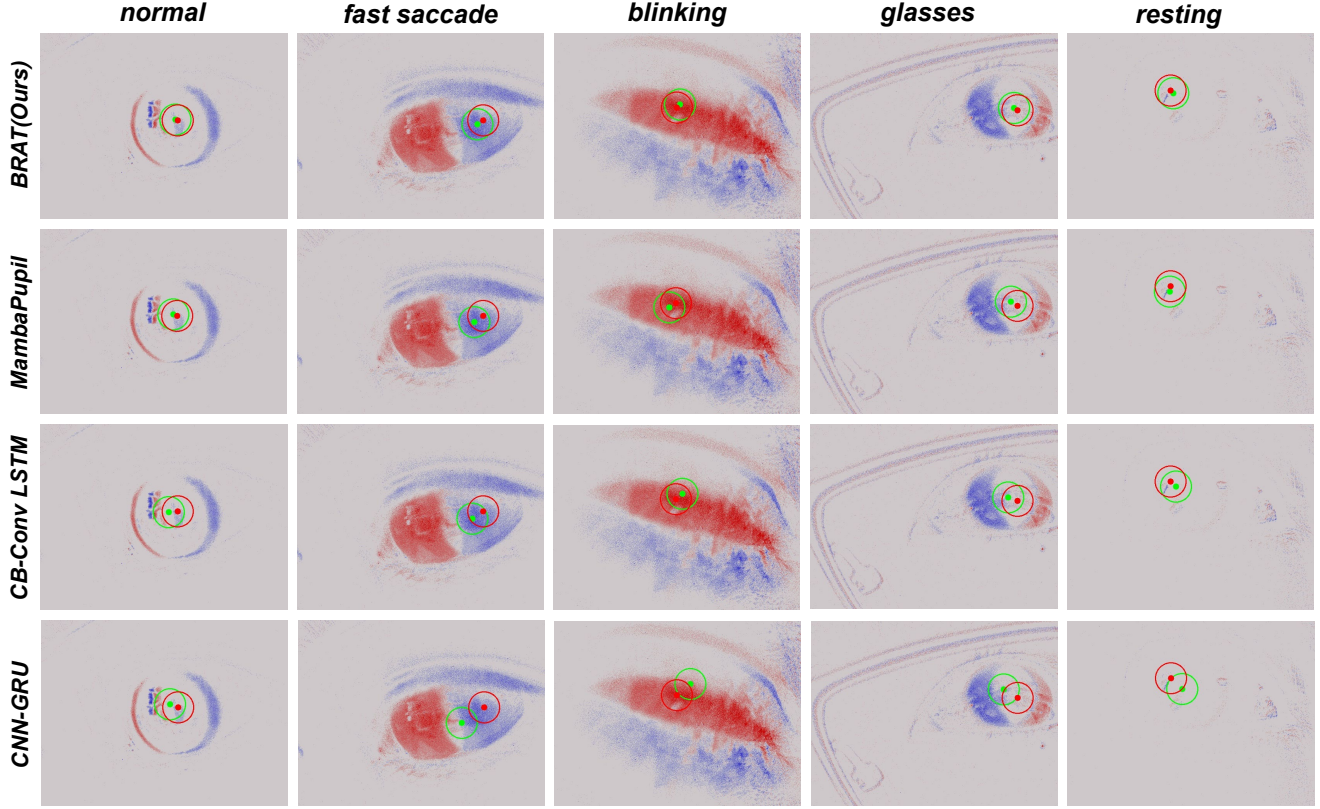


Figure 6. visual comparison of different state-of-the-art eye-tracking models and our proposed BRAT across five distinct cases. Where the dots represent the pupil positions, while the circles indicate a 5-pixel Euclidean distance error tolerance range. Specifically, the red marks denote the ground truth pupil positions, and the green marks represent the predictions of the models.

information of motion sequences. Compared to previous models such as CB-ConvLSTM, CNN-GRU, and MambaPupil, BRAT achieves more accurate and robust tracking performance.

5. Conclusion

In this paper, we propose an eye-tracking pipeline built on a transformer architecture with bidirectional positional attention (BRAT). This design enables fine-grained perception of the temporal motion characteristics of the event representation sequences. Furthermore, we introduce a multi-time-step training strategy to enhance the model’s robustness. Experimental results demonstrate that the BRAT module outperforms state-of-the-art models and achieves excellent performance on challenging samples. We hope this approach can effectively advance the application of event cameras in the field of eye-tracking. For future work, we aim to explore SNN-based architectures to extract information directly from raw event streams, fully harnessing the potential of event cameras for eye-tracking tasks.

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